PHOENIX SOLUTIONS
LET'S SHINE TOGETHER

MACHINE LEARNING WITH PYTHON

60-Days Log Sheet

Sl. No	Date	Topic Covered	Signature
1		Introduction to Machine Learning	
2		Artificial Intelligence vs Machine Learning vs Deep Learning	
3		Types of Machine Learning	
4		Machine Learning Workflow	
5		Python Basics for Machine Learning	
6		NumPy – Arrays and Numerical Computing	
7		NumPy – Array Operations and Mathematical Functions	
8		Pandas – Series and DataFrames	
9		Pandas – Data Loading and Data Manipulation	
10		Matplotlib – Basic Data Visualization	
11		Matplotlib – Advanced Charts and Customization	
12		Seaborn – Statistical Data Visualization	
13		Seaborn – Advanced Plots and Customization	
14		Setting Up ML Environment (Colab, Jupyter)	
15		Understanding Data & Dataset Structure	
16		Basic Statistics for Machine Learning	
17		Introduction to Data Visualization	
18		Case Study & Foundations Review	
19		Introduction to Data Processing	
20		Handling Missing Values	
21		Data Cleansing Techniques	
22		Feature Scaling (Normalization & Standardization)	
23		Encoding Categorical Data	
24		Handling Imbalanced Datasets (SMOTE, Near Miss)	
25		Train-Test Split & Avoiding Data Leakage	
26		Cross-Validation Techniques	
27		Feature Engineering	
28		Data Preparation Review & Practical Implementation	
29		Introduction to Scikit-learn	
30		Scikit-learn – Data Preprocessing	

Sl. No	Date	Topic Covered	Signature
31		Introduction to Supervised Learning	
32		Linear Regression	
33		Multiple & Polynomial Regression	
34		Logistic Regression	
35		Naive Bayes Classifier	
36		K-Nearest Neighbors (KNN)	
37		Support Vector Machine (SVM)	
38		Decision Trees	
39		Random Forest & Ensemble Learning	
40		Model Evaluation & Supervised Learning Review	
41		Introduction to Unsupervised Learning	
42		K-Means Clustering	
43		Hierarchical Clustering	
44		DBSCAN Clustering	
45		Dimensionality Reduction (PCA)	
46		Association Rule Learning	
47		Model Selection and Hyperparameter Tuning	
48		Grid Search and Random Search	
49		Classification Performance Metrics	
50		Confusion Matrix and Classification Report	
51		Precision, Recall and F1-Score	
52		ROC Curve and AUC	
53		Regression Model Evaluation Metrics	
54		Overfitting and Underfitting	
55		Bias-Variance Trade-off	
56		Regularization Techniques	
57		Feature Selection Techniques	
58		Pipeline and Column Transformer	
59		Practical ML Project – Data Preparation and Visualization	
60		Practical ML Project – Model Building and Evaluation	